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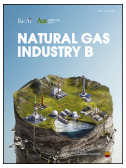

Research Article

InpAS: A sedimentary knowledge-inspired deep learning model for inpainting arbitrary strips in formation micro-imager logs

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Abstract

Formation Micro-Imager (FMI) data are critical for hydrocarbon exploration. However, variations in borehole size frequently cause gaps in FMI images, which hinder well-log interpretation. This study proposes InpAS, a conditional generative deep-learning model for image inpainting that incorporates stratigraphic constraints. InpAS integrates a customized Segment Anything Model (F-SAM) for gap detection with a diffusion network featuring a Horizontal-Cross-Vertical-Relating (HC-VR) attention module guided by Walther's Law. Ten-fold quantitative evaluation using Fréchet Inception Distance (FID), Inception Score (IS), Structural Similarity Index Measure (SSIM), RP, peak signal-to-noise ratio (PSNR), and mean absolute error (MAE) demonstrates that InpAS outperforms baseline models in reconstructing stratified FMI images. Furthermore, the reconstructed images exhibit superior retention of fine-grained visual details, textures, and geological continuity. These results demonstrates that integrating geological knowledge with geophysical logging and deep-learning algorithms significantly enhances the reliability of FMI gap-filling.
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Keywords: Deep learning; Inpainting strips; Formation Micro-imager logging

1. Introduction

The Formation Micro-Imager (FMI), developed by Schlumberger, has played an important role in hydrocarbon exploration by providing high-definition images of subsurface formations [1]. This technology facilitates diverse applications in the oil and gas sector, including borehole orientation analysis, in-situ stress direction determination, wellbore breakout monitoring, and, notably, the identification of stratigraphic features [2–6]. Compared with traditional methods, the superior image quality of FMI provides clearer insights into geological characteristics [7]. However, variations in borehole diameter can cause data gaps, known as blank strips, in the resulting images.

To address these gaps, multipoint statistics (MPS) methods, such as the FilterSim algorithm, have been employed to efficiently replicate patterns from fully imaged areas [8]. Despite their computational speed, these approaches often fail to capture the complexity of stratigraphic patterns. The high-order spatial cumulants method was subsequently introduced to improve pattern modeling across various depositional scales. Nevertheless, such methods still struggle to extract the

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latent information embedded in FMI images, which is essential for accurate reconstruction. Data-driven approaches, such as those proposed by Assous et al. [9−12], utilize curvelet transformations to decompose FMI images into multi-scale components. These methods leverage additional latent information and effectively preserve high-contrast edges for more accurate reconstructions. Furthermore, the incorporation of geometric information enhances the inpainting process. However, these techniques often fall short in extracting features from both global and local perspectives, particularly given the inherent variability of sedimentary deposits. Deep learning, particularly convolutional neural networks (CNNs), has advanced significantly in image processing [13−20]. CNN-based methods have demonstrated promise in extracting large-scale features from FMI images [21,22]. Techniques such as Partial CNN and generative adversarial networks (GANs) have further improved pattern extraction. However, these methods are constrained by the limited availability of complete images for training. To overcome this limitation, sample-free methods like the deep image prior (DIP) network have been introduced for FMI image inpainting [1,17]. While DIP reduces reliance on training samples, it may introduce artifacts and requires significant computational resources. In addition, GANs, although effective, can suffer from a collapse problem where the generator produces repetitive outputs, limiting learning [23,24].

This study introduces a novel approach that combines the strengths of geostatistical methods with deep learning. We present a model that integrates a fine-tuned Segment Anything Model (F-SAM) with a denoising diffusion probabilistic model (DDPM), enhanced by a Horizontal-Cross-Vertical-Relating (HC-VR) attention mechanism [6,25]. This model reformulates FMI image inpainting as a pixel distribution prediction problem, leveraging the ability of traditional methods to capture local details. By applying sedimentological principles, such as Walther's Law [23], we enhance image reconstruction by distinguishing between variant and invariant pixel distribution patterns. Our model significantly improves the precision of reconstructing complex electrical logging images, particularly in delineating geological contours. This results in a substantial enhancement of image detail and texture, aligning more accurately with geological contexts.

Beyond algorithmic advancements, InpAS addresses several practical challenges in hydrocarbon exploration. First, blank strips in FMI logs often interrupt dipmeter computations, forcing interpreters to manually extrapolate structural dips across incomplete intervals. InpAS restores continuous textural information, enabling automated dip picking across previously uninterpretable zones. Second, thin-bedded reservoirs, which are critical for reserve estimation, require lateral continuity of bedding signatures. Our method preserves these subtle laminated features, thereby reducing uncertainty in net-to-gross calculations. Third, the restored full-bore images serve as complete inputs for downstream machine-learning workflows, such as electrofacies classification and fracture detection, enhancing the reliability of automated geological interpretation. By providing geologically consistent reconstructions, InpAS reduces the need for manual editing and reinterpretation, saving substantial interpretation time while improving consistency across multiple wells in field development projects.

2. Motivations and methods

2.1. Principles

Fig. 1 illustrates the schematic of blank strip generation in FMI logging. Variations in borehole diameter prevent the electrode pads from fully covering the borehole wall, resulting in vertical blank strips in the acquired images. This data incompleteness necessitates the use of geological priors for image restoration.

The theoretical foundation of this approach is based on Walther's Law, a fundamental principle in sedimentology that relates vertical stratigraphic successions to lateral facies variations. This principle provides a geological prior for image

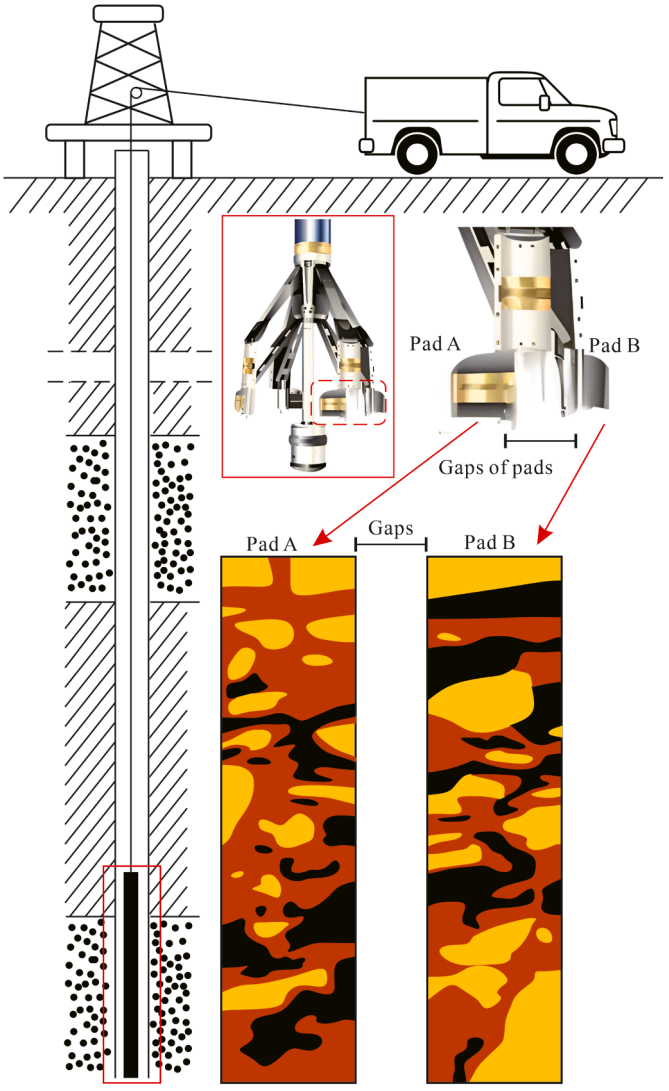

Fig. 1. Illustration of the blank strip generation process. The FMI tool image is adapted from Ref. [7].

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restoration: in conformable sedimentary sequences, the pixel distributions of vertically adjacent strata are not independent but exhibit systematic relationships reflecting the original lateral continuity of depositional environments. To formalize this relationship for FMI image inpainting, the statistical characteristics of FMI images regarding vertical and horizontal signal distributions are first analyzed.

Walther's Law states that only horizontally adjacent facies and facies zones can be vertically superimposed, provided that the sedimentary environment changes slowly and continuously [23,24]. When applied to FMI images, this law offers a novel modeling approach by using older, lower regions as references to predict the distributions of younger, upper parts. To explore the numerical representation of Walther's Law in FMI images, the distribution relationships among several local regions were analyzed. Fig. 2d−f shows the distributions from three descending local regions, each decomposable into variant and invariant components. The variant components indicate local changes in the vertical direction, reflecting subtle differences in local sedimentary structures. Conversely, the invariant components represent continuous relationships within the sediments, aligning with the principles of Walther's Law. This analysis suggests that the predictive model design should leverage information from lower regions when reconstructing adjacent upper regions.

Furthermore, Walther's Law posits that sediment evolution across horizons IS interconnected. To elucidate and compare distributions in vertical and horizontal directions, pixel distributions along these orientations were extracted (Fig. 2g and h). Using cosine similarity and signal-to-noise ratio (SNR) as metrics, variations in these distributions were evaluated. The vertical distribution (Fig. 2g) indicates that the FMI image with blank strips retains a similar overall geometry to the full FMI image, with a high cosine similarity of 0.85. However, the horizontal distribution (Fig. 2h) shows a more significant information gap, with a cosine similarity of 0.63. While the vertical direction retains more global sedimentary background information, the horizontal direction contains more detailed information regarding specific sedimentary deposit structures, as evidenced by the higher SNR in the horizontal direction (43.57 vs. 22.35).

2.2. Motivations

Building on the sedimentary constraints analyzed above, we derive a mathematical formulation that translates these geological principles into a predictive framework for image inpainting.

Given the ability of GANs to predict pixel distributions, reconstructing FMI images from a signal distribution perspective is a logical approach. However, the tendency of GANs to suffer from mode collapse necessitates a new strategy that guides deep learning models to incorporate geological and sedimentary knowledge, thereby capturing the complex geometric features of sedimentary deposits. The success of previous statistical reconstruction methods is attributed to their capacity to capture patterns in geological targets. Inspired by these methods, we revisit the task of inpainting blank strips in FMI logging data. Fig. 2a presents a complete FMI image, whereas Fig. 2b shows a simulated FMI image with artificially introduced blank strips. The stratigraphic succession in these images progresses from older deposits at the bottom to younger ones at the top. Fig. 2c compares the pixel distributions of the full and defective FMI images, highlighting their similar overall geometries. Notably, the pixel distribution of the full FMI image contains specific, localized signals that are absent in the defective image. This observation suggests that the primary challenge in inpainting is accurately modeling these missing local signals. To address this, we predict and reintroduce the deficient local signals X_m to the remaining signals X_n, thus reconstructing the full distribution X_w. This task can be mathematically expressed as:

$$X_w = X_m + X_n = \Phi(X_n) + X_n \tag{1}$$

where $\Phi(\cdot)$ denotes the predictive function.

Latent information from fully imaged local regions, combined with geometric cues from sedimentary deposits, is critical for inpainting blank strips, as demonstrated in Refs. [7,8,26−28]. This insight aligns closely with the sedimentary interpretation practices of well-logging engineers. The significance of local sedimentary data in deciphering complex geological patterns suggests that modeling methods grounded in sedimentary principles can address the limitations of current deep learning models. Specifically, sedimentary knowledge serves as valuable prior information in model design. By integrating sedimentological principles with image signal reconstruction, deep learning models can achieve superior FMI image reconstruction. Inspired by Walther's Law, we propose an improved formulation:

$$X_w = \sum \Phi(X_{n,i}, X_{n,i-k}) + X_n \tag{2}$$

In this equation, i indexes the regions segmented from bottom to top, and k is a hyperparameter controlling the influence distance of adjacent regions, consistent with Walther's Law.

This yields a refined formulation that integrates both vertical and horizontal information:

$$X_w = \sum \Phi\left(\mathrm{Y}\left(X_{n,i,j}, X_{n,i-k,j-k}\right)\right) + X_n \tag{3}$$

where Y is a mechanism integrating vertical and horizontal information, while j indexes regions segmented from left to right. Specifically, Walther's Law posits that vertical facies successions reflect lateral environmental shifts, implying that younger deposits are predictable only from immediately underlying (older) strata that were horizontally contiguous during deposition. We translate this principle into mathematical constraints via the i-k indexing scheme in Equation (2), where k denotes the vertical distance to underlying regions. This formulation ensures that the reconstruction of region i (younger strata) depends strictly on regions i-k (older strata), thereby mathematically preventing information flow from younger to older deposits. Furthermore, the horizontal index j and the integration function Y in Equation (3) explicitly model the lateral facies relationships (horizontal adjacency) required

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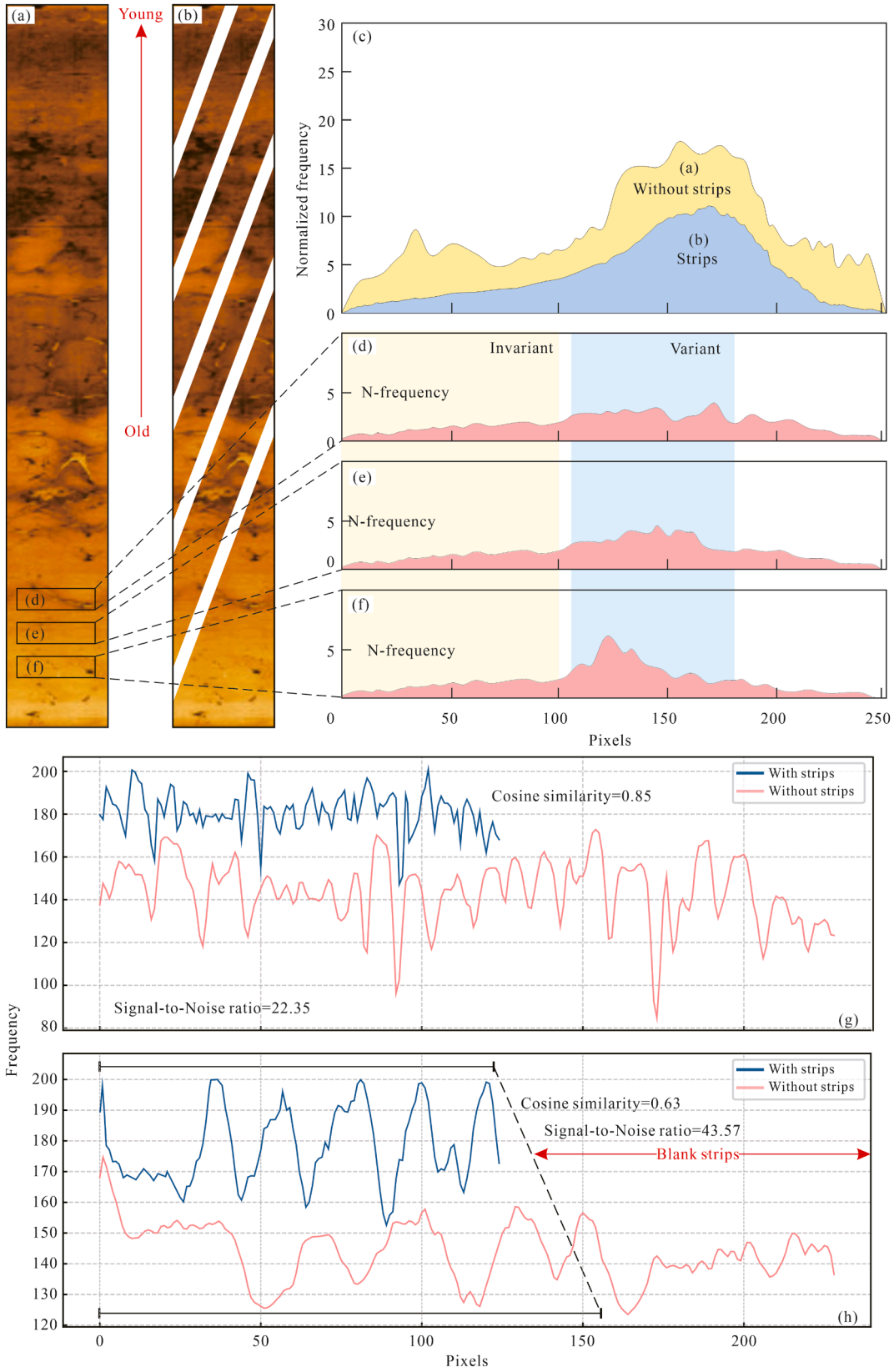

Fig. 2. Experimental generation of blank strips. (a) Full FMI image; (b) Image from (a) with generated blank strips; (c) Visualization of pixel distributions for (a) and (b); (e) and (f) pixel distributions of local windows; (g) and (h) pixel distributions in vertical and horizontal directions.

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by Walther's Law, while the vertical component captures the invariant continuity of sedimentation. Thus, the predictive function Φ operates as a stratigraphic autoregressive model in which geological time (bottom-to-top) dictates the direction of computational dependency.

As illustrated in Fig. 3, the InpAS framework implements a conditional generation pipeline comprising two algorithmic stages. Initially, the input FMI image X_{FMI} is processed by the F-Segment Anything Model (SAM) model Φ_1 using point and text prompts to precisely localize blank strips and generate a binary mask X_{blank} (Equation (4)). Subsequently, this mask and the partially observed image serve as conditioning inputs to the HC-VR DDPM, which executes the reverse denoising process (Equations (7) and (8)) guided by this conditioning. Specifically, local windows of size 20 × 20 pixels traverse the image from bottom to top. Within each window, the HC-VR mechanism (Equations (9)–(11)) applies vertical causal attention, enforcing Walther's Law and horizontal facies correlation, to reconstruct the missing pixel distributions, thereby yielding the final inpainted image X_w. Sedimentary knowledge, rigorously represented mathematically as $\Upsilon(X_{n,\,i,\,j},\, X_{n,i-k,\,j-k})$, informed the design of the HC-VR Attention block (Horizontal-Cross-Vertical-Relating Attention). This block, combined with the initial detection and masking of blank strips via the SAM model, guides the subsequent diffusion model to generate the required FMI images.

2.3. Fine-tuned Segment Anything Model (F-SAM)

Directly applying deep learning models for global image reconstruction, while ignoring the spatial associations of geological targets, may result in the loss of depositional information in areas outside blank strips. Therefore, reconstruction should be restricted to regions adjacent to blank strips rather than applied globally. Although classical digital image processing techniques (e.g., Otsu thresholding, Canny edge detection) suffice for detecting idealized zero-value strips in clean boreholes, they produce hard binary masks decoupled from downstream processing and often fail to handle gradual intensity transitions caused by borehole rugosity or partial electrode contact. As a foundation model in computer vision, SAM has demonstrated remarkable performance in segmentation tasks, particularly in zero-shot scenarios [29]. This capability shows great potential for the rapid and accurate localization of strips. However, SAM's performance on specific segmentation tasks is suboptimal, primarily because its training on common natural images limits its adaptability to specialized imagery. To preserve SAM's zero-shot generalization capability while enhancing fine-tuning flexibility, this study adapts the High-Quality SAM (HQ-SAM) model [30] by incorporating additional adaptor layers. These adaptor layers are integrated into the original image encoder, prompt encoder, and prompt decoder, and remain learnable during fine-tuning. Fig. 4 illustrates the application of the fine-tuned SAM (F-SAM, denoted as $\phi_1 \in \Phi$) to the task of detecting blank strips in FMI images. The input FMI images containing blank strips (X_{FMI}) are processed through Vision Transformer (ViT) blocks, as depicted in Fig. 4. Parameters within the layer normalization, multi-head attention, and multi-layer perceptron modules are frozen (indicated by snow symbols in Fig. 4), whereas the adaptor layers are set as learnable to enable the localization of blank strips (indicated by fire symbols in Fig. 4). By leveraging point and text prompts integrated by the prompt encoder, the segmentation of blank strips (X_{blank}) is achieved. This process is mathematically articulated as:

$$X_{\text{blank}} = \phi_1(\theta, X_{FMI}, X_{\text{Point}}, X_{\text{Text}}) \tag{4}$$

where X_{Point} and X_{Text} denote the point and text prompts, respectively.

2.4. Diffusion model with HC-VR attention

Formulating inpainting as a distribution prediction problem necessitates a flexible deep learning model capable of reconstructing subtle variations in pixel distributions. Diffusion models excel at capturing intricate image details, as the iterative steps in the distribution projection process facilitate diverse variation transformations. This capability makes

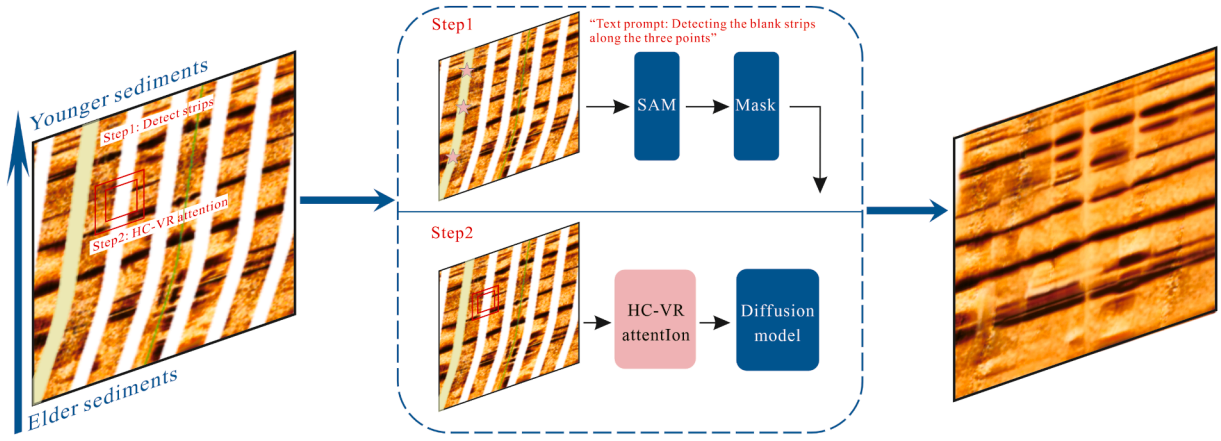

Fig. 3. Workflow of InpAS. Initially, text prompts are combined with SAM to identify and mask blank strips. Subsequently, a sedimentary knowledge-guided diffusion model is employed to inpaint these regions.

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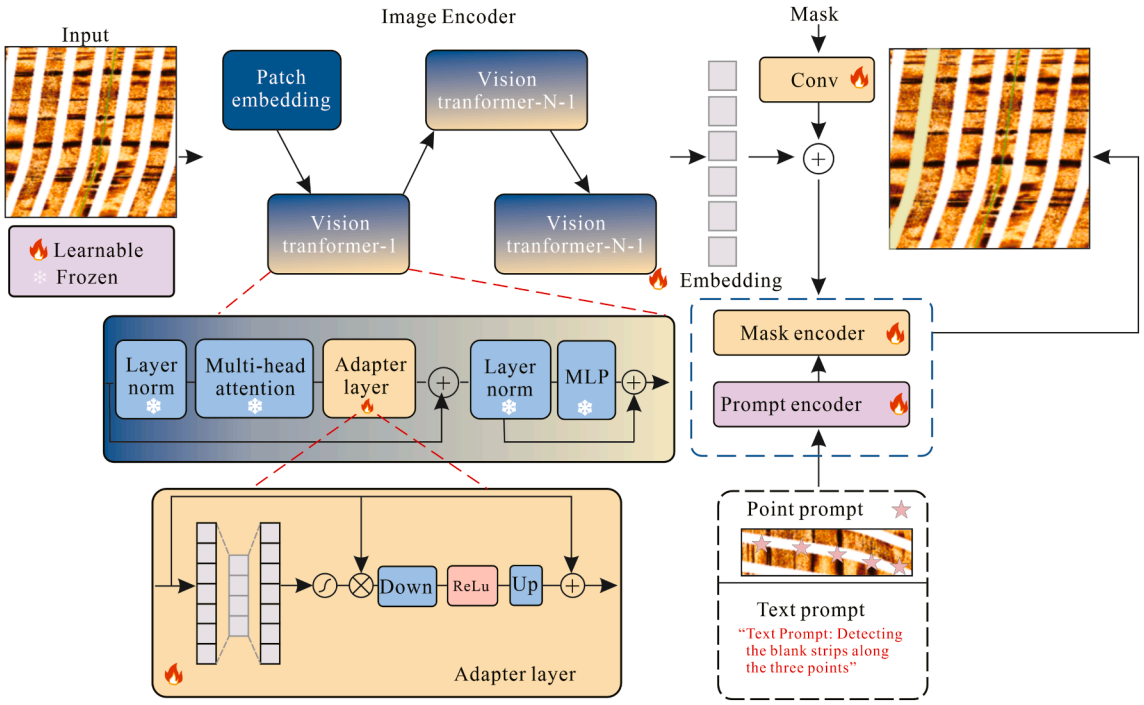

Fig. 4. Schematic of the fine-tuned Segment Anything Model (F-SAM) employed in this study to detect blank strips in FMI images.

diffusion models particularly suitable for inpainting tasks involving blank strips.

The diffusion model comprises two processes: the forward process q and the reverse process p. The forward process incrementally adds Gaussian noise β to the original FMI images and corresponding masks (X_{FMI}, X_{masks}) generated by the SAM model across iterative time steps t, until the data structure is completely obscured. Conversely, the reverse process systematically removes noise via a denoising function, ultimately restoring the original data X_{FMI}. The forward processes are mathematically defined as follows:

$$q(x_{0:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}) \tag{5}$$

$$q(x_t|x_{t-1}) \sim N\left(x_t; \sqrt{1-\beta_t}\,x_{t-1}, \beta_t I\right) \tag{6}$$

where $x_t \in R_t$ denotes the data at time step t. Equation (5) characterizes the probability distribution of the sequence $x_{0:T}$ conditioned on the initial state x_0. As indicated by Equation (6) x_{t-1}, x_t follows a normal distribution with mean $\sqrt{1-\beta_t}x_{t-1}$ and variance $\beta_t I$, given x_{t-1}. The reverse process is defined as:

$$P_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p\theta(x_{t-1}|x_t) \tag{7}$$

$$P_\theta(x_{t-1}|x_t) \sim N\left(x_{t-1}; \mu_\theta(xt,t), \sigma_t^2 I\right) \tag{8}$$

In these equations, Equation (7) defines the probability distribution of the sequence $x_{0:T}$ under parameter θ, encompassing the probability of the final state x_T and the conditional probabilities from x_t to x_{t-1} at each time step t. Equation (8) specifies that, given x_t, the probability distribution of x_{t-1} follows a normal distribution with mean $\mu_\theta(x_t,t)$ and variance $\sigma_t^2 I$.

Although denoising diffusion probabilistic models (DDPMs) effectively model complex data distributions through processes p and q, the stochastic nature of noise addition and removal can cause deviations from the specific distributions associated with sedimentary deposits. To address this issue, we introduce the HC-VR Attention mechanism, which integrates vertical and horizontal attentions to guide DDPMs in more accurately capturing distributions influenced by sedimentary patterns.

The integration function Y in Equation (3) is formally defined as a cross-directional attention fusion operator that combines vertical stratigraphic continuity with horizontal facies variation information. Given local region features $X_{n,\,i,j}$ (current position) and $X_{n,i-k,j-k}$ (underlying adjacent position), the function is implemented as:

$$\mathrm{Y}\left(X_{n,i,j}, X_{n,i-k,j-k}\right) = LayerNorm\left(W_o \cdot \mathrm{Concat}\left[\mathrm{Attn}_v\left(X_{n,i,j}\right), \mathrm{Attn}_h\left(X_{n,i-k,j-k}\right)\right]\right) \tag{9}$$

where $Attn_v$ and $Attn_h$ denote multi-head self-attention operations along the vertical and horizontal axes, respectively, and $W_o \in \mathbf{R}^{\mathbf{d}\times\mathbf{2d}}$ is a learnable projection matrix. Vertical attention incorporates a geological causal mask to enforce Walther's Law, ensuring that the reconstruction of upper (younger) strata attends only to lower (older) strata:

$$\mathrm{Attn}_v(X) = \mathrm{Softmax}\left(\frac{Q_v K_v^T}{\sqrt{d_k}} + M_{geo}\right) \tag{10}$$

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where Q_v, K_v, and V_v denote query, key, and value matrices projected from the input features; d_k represents the dimension of the key vectors; and M_{geo} denotes a causal mask that assigns -∞ to positions above the current stratum (preventing attention to younger deposits) and 0 to positions below. This mathematically implements the principle that only horizontally adjacent facies can be vertically superimposed. Horizontal attention captures lateral facies relationships without directional constraints:

$$\mathrm{Attn}_h(X) = \mathrm{Softmax}\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right) V_h \tag{11}$$

In practice, the HC-VR mechanism is implemented as residual blocks within the U-Net backbone of the DDPM, operating on feature patches of size 20 × 20 pixels (as determined in Section 4.1). The complete algorithmic procedure, including tensor dimension specifications and forward pass implementation, is provided in Algorithm 1 to ensure full reproducibility.

Algorithm 1. The pseudocode of the HC-VR attention mechanism

Algorithm HC-VR Attention Mechanism ($X \in R^{B\times C\times H\times W}$, M_{geo})

Input: Feature map X ∈ $R^{B\times C\times H\times W}$, geological mask M_{geo}
Output: Enhanced feature map X_{out} ∈ $R^{B\times C\times H\times W}$
Vertical Attention (Stratigraphic Continuity)
Q_v, K_v, V_v ← **Linear$_v$**(X) # Project to d_k dimensions
A_v ← $Q_v \times K_v^T$/**Sqrt** (d_k) # attention scores
A_v ← $A_v + M_{geo}$ # Apply causal mask (Walther's Law)
$Attn_v$ ← **Softmax** (A_v) × V_v
Horizontal attention (Facies Variation)
Q_h, K_h, V_h ← **Linear$_h$**(X)
A_h ← $Q_h \times K_h^T$/Sqrt (d_k)
$Attn_h$ ← **Softmax** (A_h) × V_h
Cross-Directional Fusion (Equation (9))
Concat ← [$Attn_v$; $Attn_h$] # Concatenate along channel dim
X_{out} ← **LayerNorm** (**MLP**(Concat)) + X # Residual connection
return X_{out}

2.5. Geological constraints in network architecture

The causal mask M_{geo} in Equation (10) implements Walther's Law within the attention mechanism. By assigning $-\infty$ to attention scores for positions above the current stratum (future/younger deposits) and 0 to those below (past/older deposits), the mask restricts the model to attend only to underlying geological information when reconstructing blank strips. This directional constraint ensures that the diffusion model respects the chronological order of sedimentation during the reverse denoising process, thereby transforming the geological principle that "the present is the key to the past" into a hard mathematical boundary condition. The integration of vertical causal attention (stratigraphic continuity) and horizontal bidirectional attention (lateral facies variation) rigorously operationalizes Walther's Law within the deep learning framework.

3. Experiments and results

3.1. Datasets

Seven wells with high-quality FMI data were selected, comprising 184 m of FMI logs. The training data for fine-tuning the F-SAM model were annotated in-house. Acquiring sufficient full-borehole images presents a significant challenge for training deep learning models. For the HC-VR DDPMs, training data were constructed using a self-supervised approach wherein synthetic masks simulating electrode exclusion zones were generated on existing valid FMI regions (areas with intact electrode coverage). Specifically, a random mask method was employed to generate these data. In addition, to enhance the gap-filling performance of the HC-VR DDPMs, this random mask method was applied to large datasets, including Place 2, Places Challenge, ImageNet, Open Images, CIFAR-10, and CIFAR-100. The large and base HC-VR DDPM models were trained to facilitate subsequent transfer learning.

3.2. Experimental setup

The experiments in this study were conducted on a Python-based online platform (AutoDL, https://www.autodl.com/), with the network training and testing performed on an NVIDIA A100 GPU. In the first step, involving blank strip detection, a momentum-based Adam optimizer was employed to fine-tune SAM models. The parameters included T values of 1, 2, and 4, with corresponding momentum values of 0.1, 0.5, and 0.9. The initial learning rate β was set to 2×10^{-4}, adjusted via an exponential decay function with a rate of 0.999. The batch size was 32, and the number of iterations was 3000. In the second step, which involved inpainting blank strips, training parameters remained consistent with the first step, with the number of iterations set to 200. All reconstruction methods were evaluated using 10-fold cross-validation, with quantitative results reported as mean ± standard deviation.

3.3. Evaluation metrics

In this study, recall and Intersection over Union (IoU) [31] were employed to evaluate the performance of the F-SAM model in detecting blank strips. For the inpainting task, Fréchet Inception Distance (FID) [32], Inception Score (IS) [33,34], Structural Similarity Index Measure (SSIM) [35], recall-precision (RP), peak signal-to-noise ratio (PSNR), and mean absolute error (MAE) were utilized to provide complementary assessments at the distribution, structure, signal, feature, and pixel levels. Lower FID and MAE values indicate superior performance, whereas higher IS, SSIM, RP, and PSNR values denote better quality. All reconstruction results were evaluated using a 10-fold cross-validation protocol and are reported as mean ± standard deviation. To ensure that the assessment accurately reflects inpainting quality, all metrics were calculated exclusively within the masked blank-strip

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regions rather than over the entire images. Using the binary mask M generated by F-SAM, metrics were computed as Metriclocal = Metric (Igen × M, Ireal × M). This approach prevents large unaltered areas from dominating the scores. FID was computed using the CLIP image encoder to extract features from the generated and reference images.

$$\mathrm{FID} = \left\|\mu_r - \mu_g\right\| + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^2\right)^{\frac{1}{2}} \tag{12}$$

where μ_r, μ_g, Σ_r, and Σ_g denote the means and covariances of the real and generated images, respectively. The IS another critical metric for evaluating GAN performance, and IS calculated as follows:

$$\mathrm{IS} = \exp\, \mathbb{E}_{x\sim p_g}\, D_{\mathrm{KL}}(p(d|x)\|p(y)) \tag{13}$$

In this equation, $p(d|x)$ and $p(y)$ denote the conditional probability of the predicted label y given the input image x and the marginal probability of the label y, respectively. A higher IS value indicates superior quality and diversity in the generated images. The SSIM is an image quality assessment metric that aligns more closely with human visual perception than pixel-based error measures by accounting for luminance, contrast, and structural similarities. Its mathematical formulation IS expressed as:

$$SSIM(x,y) = \frac{\left(2\mu_x\mu_y + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{14}$$

where x and y denote the grayscale representations of the images under comparison, while μ_x, μ_y, σ_x^2, σ_y^2, and σ_{xy} represent the means, variances, and covariance, respectively. The constants c_1 and c_2, typically defined as $(k_1L)^2$ and $(k_2L)^2$ (where L is the maximum grayscale level and k_1, k_2 are small constants), ensure numerical stability during computation. The SSIM value ranges from −1 to 1, with values approaching 1 indicating a higher degree of structural similarity.

3.4. Results

Quantitative accuracy and field-data plausibility were evaluated separately. For the controlled accuracy assessment, complete FMI intervals were selected, and blank strips were artificially introduced, with the corresponding unmasked images serving as reference targets. This design facilitates direct comparison between reconstructed and observed pixels within the masked regions. For naturally incomplete FMI images, fully observed ground truth is generally unavailable because gaps between electrode pads are inherent to FMI acquisition. Consequently, the results from these field examples are interpreted as evidence of statistical consistency and geological plausibility rather than absolute pixel-level accuracy. The global pixel distributions shown in Fig. 2c illustrate the recovery of local signal components missing from the defective images.

The detection results for blank strips are illustrated in Fig. 5. By utilizing point prompts, the F-SAM model successfully segments and generates masks for the blank strips, achieving an IoU score exceeding 0.9 for each instance, which demonstrates its exceptional localization capability. Furthermore, twelve deep learning models were fine-tuned for blank strip detection. As shown in Fig. 6, the SAM model outperformed the others; however, it possesses a significantly larger number of parameters. Considering both efficiency and accuracy, the proposed F-SAM model offers a balanced solution with fewer learnable parameters (4.3 M) and satisfactory accuracy (IoU of 0.87 compared to 0.92), as depicted in Fig. 6. This indicates that the F-SAM model incurs only a 5% loss in IoU while reducing the parameter count by 98.7%.

To visually assess the performance of the InpAS model, Fig. 7 shows a comparative visualization of the generated images. The first four columns of Fig. 7 show representative images generated from the dataset using InpAS. Although the generated images exhibit somewhat blurry boundaries between geological features and the background, suggesting potential for resolution improvement, they show clearer outlines. The model effectively captures subtle details and textures, particularly the intricate framework of sedimentary deposits, such as their lamellar and clot structures.

The IS-FID cross-plot in Fig. 8 and the quantitative comparison in Table 1 evaluate InpAS against representative multipoint-statistics, sparse-reconstruction, CNN-, GAN-, score-, and diffusion-based methods. Under the 10-fold evaluation protocol, InpAS achieved the best mean results across all six reconstruction metrics: FID = 4.36 ± 0.22, IS = 11.29 ± 0.77, SSIM = 48.60 ± 0.84%, RP = 19.60 ± 0.62%, PSNR = 49.78 ± 5.83 dB, and MAE = 4.57 ± 0.10. Compared with the closest baseline, DDPM+, InpAS reduced FID by 18.66%, increased IS by 18.34%, increased RP by 2.34 percentage points, improved PSNR by 1.58 dB, and reduced MAE by 0.10. These results support improved performance under the present dataset, masking protocol, and stratified FMI conditions, rather than universal superiority across all geological settings. These results, together with the PCA in Fig. 9 showing that the reconstructed images preserve multidimensional geological feature distributions, with relative deviations of cluster centers below 0.5, support improved performance under the present dataset, masking protocol, and stratified FMI conditions, rather than universal superiority across all geological settings.

In addition, the performance of the InpAS model was evaluated using a collection of FMI images without blank strips. Fig. 9a displays the PCA decomposition results for the original FMI images. After manually introducing blank strips and applying inpainting with the InpAS model, the decomposition results for the reconstructed images are shown in Fig. 9b. The inpainted FMI images exhibit a distribution closely resembling that of the raw images. Moreover, the deviation of the clustering center points was calculated, with relative deviations illustrated in Fig. 9c. The relative deviations of the clustering center points were less than 0.5, indicating a high degree of consistency between the original and inpainted FMI images.

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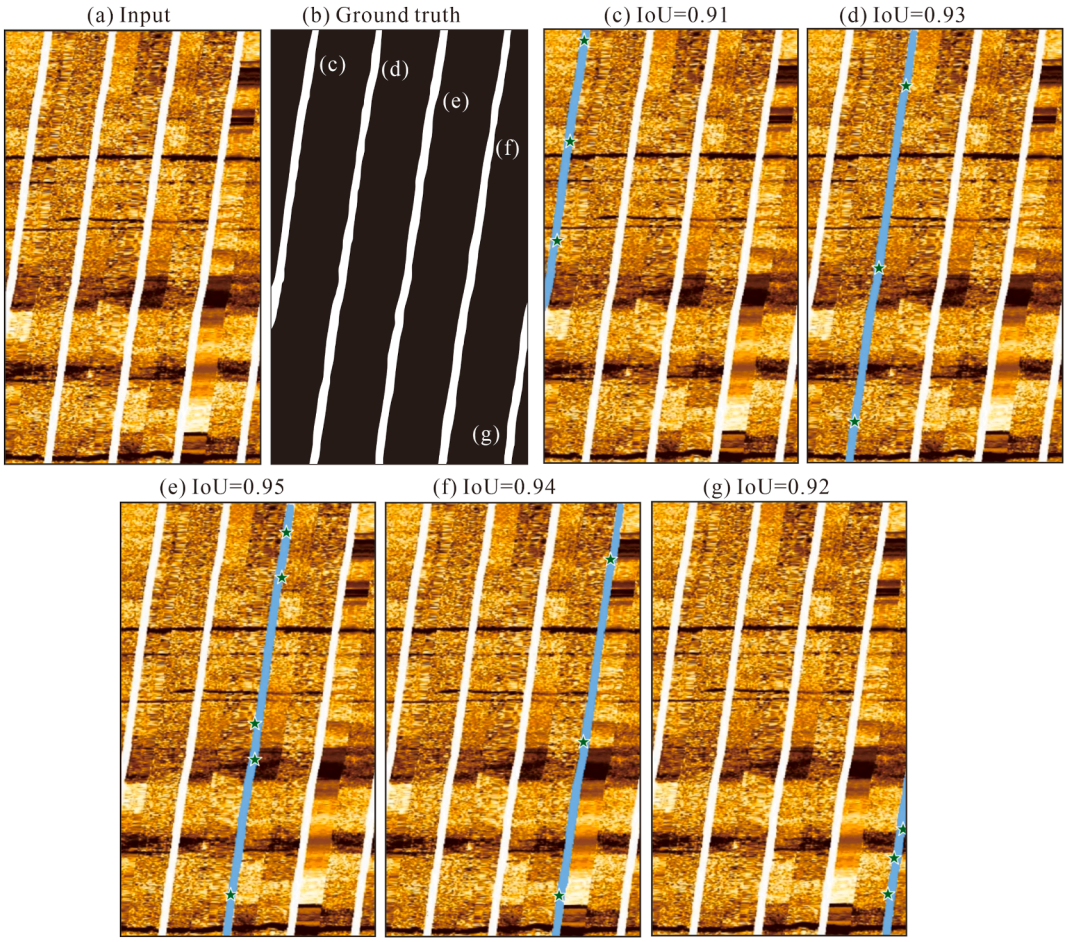

Fig. 5. Detection of blank strips using the F-SAM model.

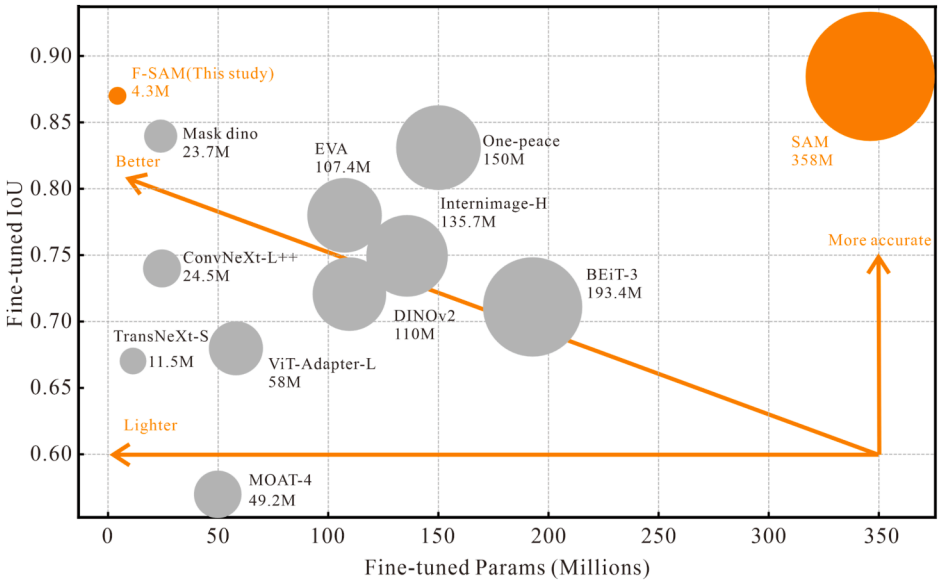

Fig. 6. Comparison of IoU versus learnable parameters for different models.

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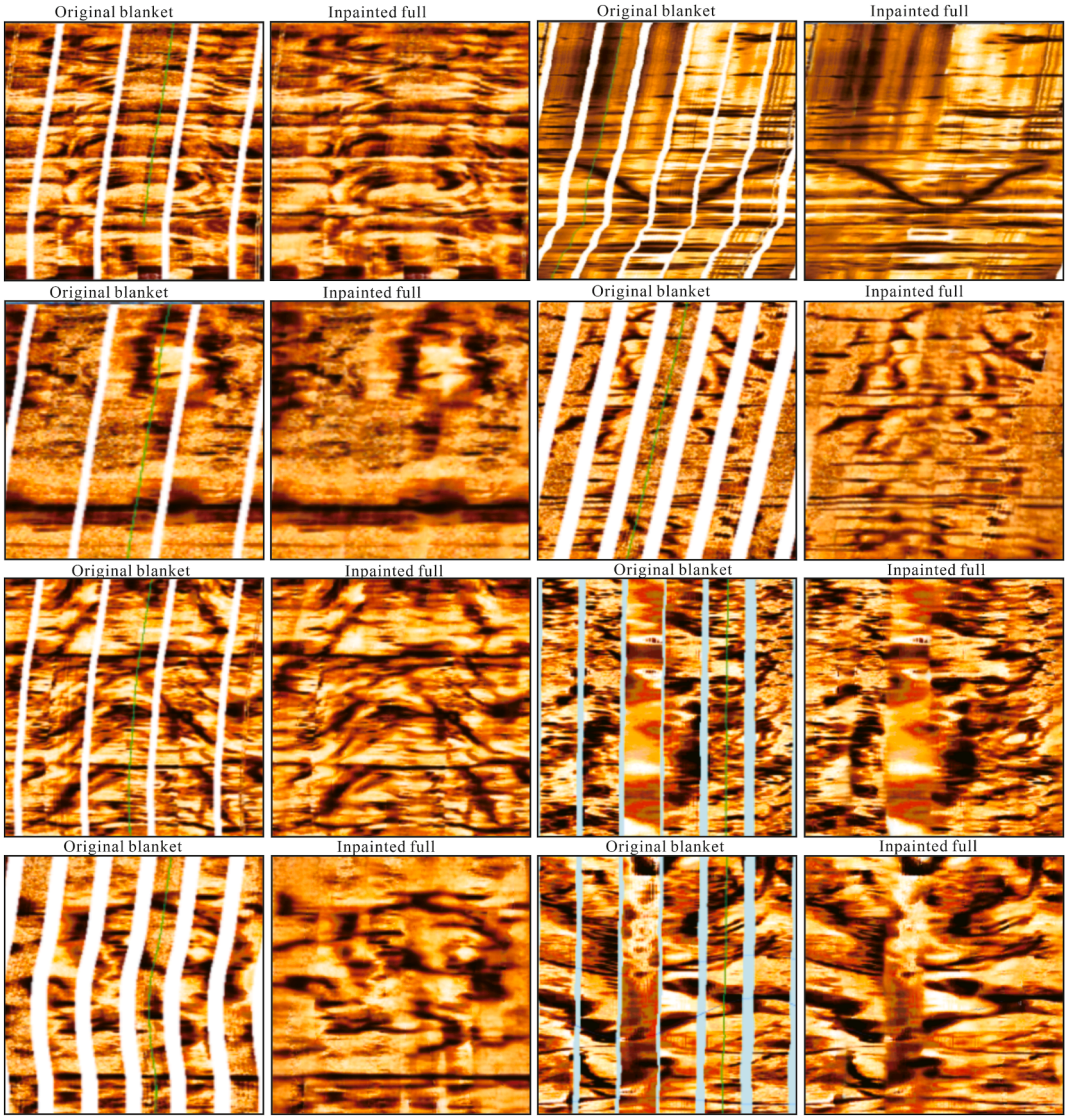

Fig. 7. Inpainted FMI blank strips using the InpAS model.

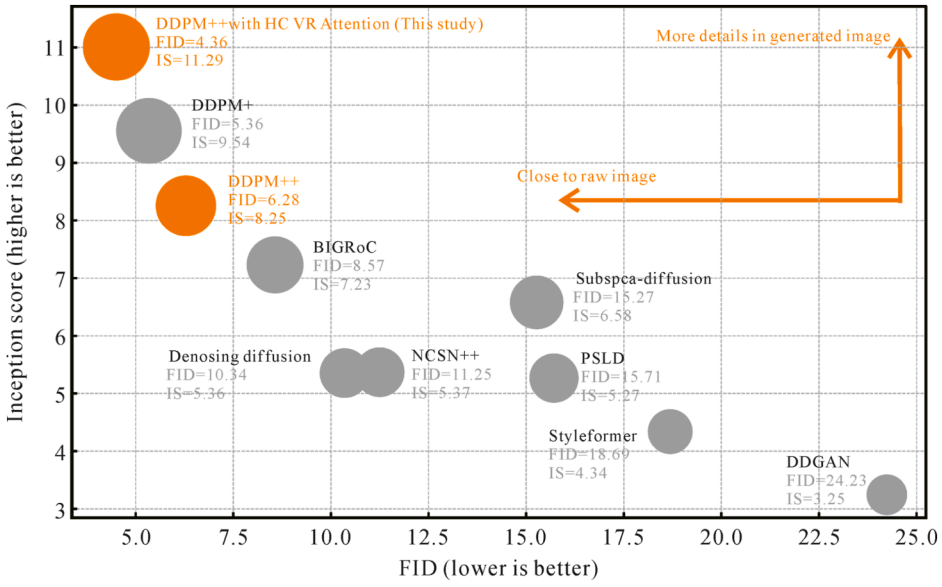

Fig. 8. Comparison of inception scores and FID across different models.

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Table 1
Quantitative comparison of InpAS with representative FMI blank-strip reconstruction and generative baseline methods.

Method	FID (lower)	IS (higher)	SSIM (%) (higher)	RP (%) (higher)	PSNR (dB) (higher)	MAE (lower)
InpAS (DDPM + HC-VR)	**4.36 ± 0.22**	**11.29 ± 0.77**	**48.60 ± 0.84**	**19.60 ± 0.62**	**49.78 ± 5.83**	**4.57 ± 0.10**
DDPM+	5.36 ± 0.57	9.54 ± 0.96	48.40 ± 0.80	17.26 ± 0.36	48.20 ± 7.07	4.67 ± 0.14
DDPM++	6.28 ± 0.34	8.25 ± 0.79	47.50 ± 0.55	15.16 ± 0.70	47.87 ± 7.20	4.73 ± 0.50
BIGRoC	8.57 ± 0.38	7.23 ± 0.29	43.20 ± 0.51	14.46 ± 0.26	41.17 ± 5.63	5.77 ± 0.91
Denoising Diffusion	10.34 ± 0.47	5.36 ± 0.56	38.80 ± 0.48	12.96 ± 0.13	40.10 ± 8.83	7.70 ± 0.58
NCSN++	11.25 ± 0.44	5.37 ± 0.61	38.10 ± 0.27	12.86 ± 0.43	36.74 ± 3.77	8.57 ± 0.14
Subspca-Diffusion	15.27 ± 0.53	6.58 ± 0.23	36.30 ± 0.32	12.46 ± 0.30	34.49 ± 7.40	9.43 ± 0.36
PSLD	15.71 ± 0.29	5.27 ± 0.34	25.10 ± 0.32	11.36 ± 0.79	30.12 ± 7.43	9.70 ± 0.80
Styleformer	18.69 ± 0.90	4.34 ± 0.38	24.50 ± 0.57	10.26 ± 0.19	26.15 ± 6.17	9.93 ± 0.84
DDGAN	24.23 ± 1.00	3.25 ± 0.82	22.90 ± 0.50	10.16 ± 0.18	23.41 ± 3.77	12.17 ± 0.22
FilterSim [9,10]	54.57 ± 0.93	5.90 ± 1.00	21.30 ± 0.58	9.28 ± 0.65	23.35 ± 3.87	13.63 ± 0.31
Direct sampling [11]	34.21 ± 0.92	3.76 ± 0.36	17.00 ± 0.56	8.80 ± 0.77	20.57 ± 5.53	14.20 ± 0.36
Higher-order spatial cumulants [12]	45.21 ± 0.23	3.74 ± 0.57	15.90 ± 0.82	4.86 ± 0.78	19.75 ± 5.57	14.27 ± 0.68
Curvelet-based reconstruction [13,14]	57.52 ± 0.27	2.36 ± 0.52	15.40 ± 0.78	4.42 ± 0.33	19.60 ± 5.83	15.50 ± 0.59
Partial convolution [31]	34.52 ± 0.52	4.10 ± 0.95	14.80 ± 0.38	3.50 ± 0.83	15.81 ± 8.73	16.30 ± 0.83
DIP [1,24]	40.25 ± 0.27	2.28 ± 0.16	14.30 ± 0.65	2.68 ± 0.22	14.15 ± 6.63	16.43 ± 0.93
GAN-based inpainting [26]	28.52 ± 0.76	2.32 ± 0.42	13.90 ± 0.99	2.66 ± 0.92	13.66 ± 8.07	16.47 ± 0.84

Note: Values represent the mean ± standard deviation from a 10-fold evaluation. SSIM and RP are expressed as percentages. All metrics were calculated exclusively within the masked blank-strip regions. Bold indicates the best observed result.

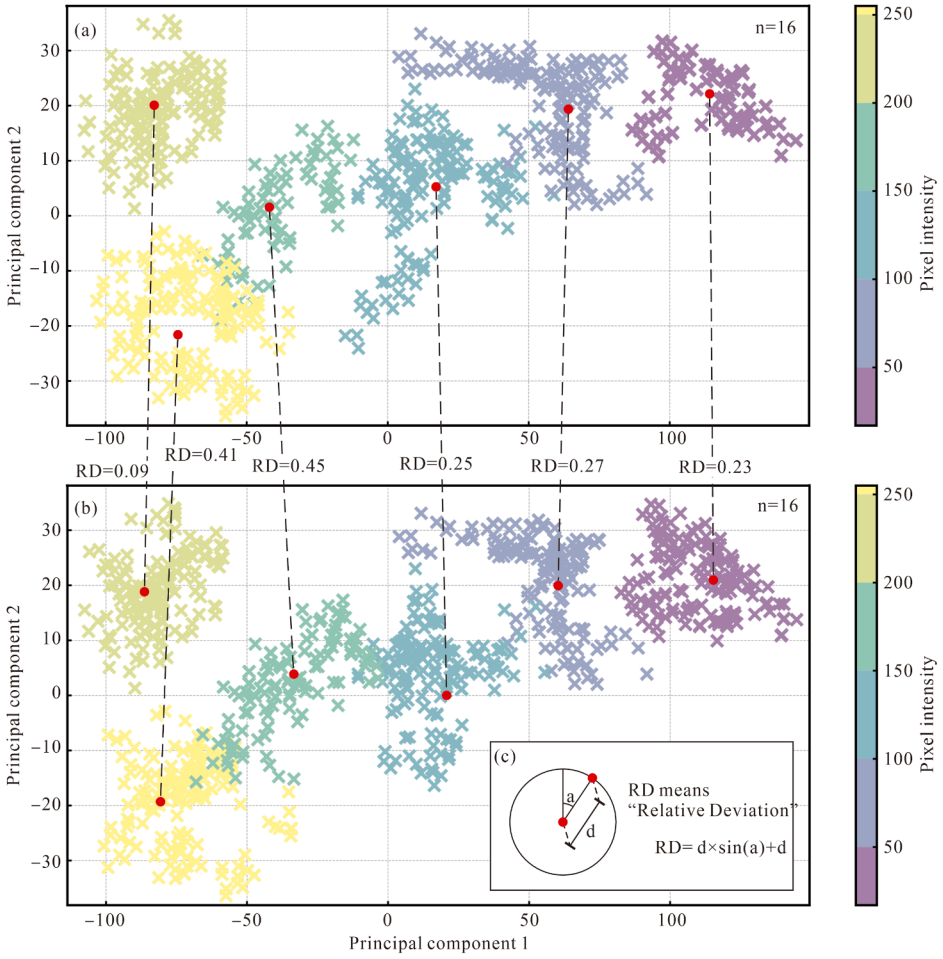

Fig. 9. PCA decomposition and pattern matching of original and inpainted FMI images.

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4. Discussion

4.1. Model robustness and deployment considerations

Beyond the parametric sensitivities discussed in subsequent sections, several intrinsic characteristics of the InpAS architecture warrant explicit discussion.

First, the HC-VR attention mechanism introduces a trade-off between geological fidelity and computational cost. The vertical causal mask, while essential for enforcing Walther's Law, increases the effective receptive field along the stratigraphic direction. Consequently, model performance depends on the correct identification of the younging direction. In our experiments, inadvertently orienting the input image with younger strata at the bottom (i.e., reversed stratigraphic polarity) reduced the SSIM by approximately 12%. This finding confirms that the geological prior functions not as a passive regularization but as an active constraint requiring proper data preparation.

Second, the DDPM iterative denoising schedule, defined by the timestep T, affects both reconstruction quality and inference speed. We observed that $T = 1000$ provides the highest textural fidelity (FID = 4.36), whereas reducing T to 500 increases the FID to 6.12 but accelerates inference by nearly 40%. For routine well-site applications prioritizing rapid feedback, $T = 500$ offers a practical compromise. The DDPM U-Net backbone, comprising approximately 86 million parameters, operates stably on 20×20 pixel patches. However, for larger images exceeding 500×500 pixels, we recommend dividing the image into overlapping patches and averaging predictions to prevent memory overflow on standard GPUs, such as NVIDIA A100 with 40 GB memory.

Third, although the F-SAM detection module achieves high accuracy (IoU = 0.87), it introduces a non-negligible latency of approximately 1.2 s per 250×250 image patch, which may limit its use for real-time logging interpretation. As discussed in Section 4.5, we provide a lightweight alternative based on Otsu thresholding, which reduces detection time to approximately 0.06 s per patch with only a marginal IoU decrease from 0.87 to 0.83 under ideal borehole conditions. We recommend selecting the detection strategy based on borehole quality: Otsu for stable, conductive mud environments with sharp strip edges, and F-SAM for rugose or washed-out boreholes where strip boundaries are diffuse.

Fourth, the model's generalization to unseen lithologies remains an open question. Although pretraining on diverse natural image datasets (ImageNet, Place 2) provides a broad feature basis, fine-tuning was performed exclusively on carbonate-dominated FMI logs. When applied to a small set of sandstone FMI images excluded from the training set, the FID increased to 8.73, indicating performance degradation. This suggests that transfer learning or domain adaptation is necessary before deploying InpAS in clastic or mixed-lithology settings. We emphasize that the current model is optimized for stratified carbonate reservoirs; users targeting other lithologies should interpret results with appropriate caution.

4.2. Selection of an appropriate window for DDPM

The DDPM inpainting process operates within both pre-set and moving windows. Window size, a critical hyperparameter, significantly influences inpainting quality.

As illustrated in Fig. 10, varying window sizes yield different inpainting outcomes. The average width of blank strips is 8 pixels, while the average inter-strip distance is 49 pixels (Fig. 10a). When the window size is smaller than both the strip width and the inter-strip distance (Fig. 10b), the window fails to capture complete blank regions due to insufficient adjacent information for inpainting guidance. The minimal repair observed in Fig. 10b confirms that the window size must exceed the strip width. As the window size increases beyond the strip width but remains significantly smaller than the inter-strip distance (Fig. 10c–e), the window consistently

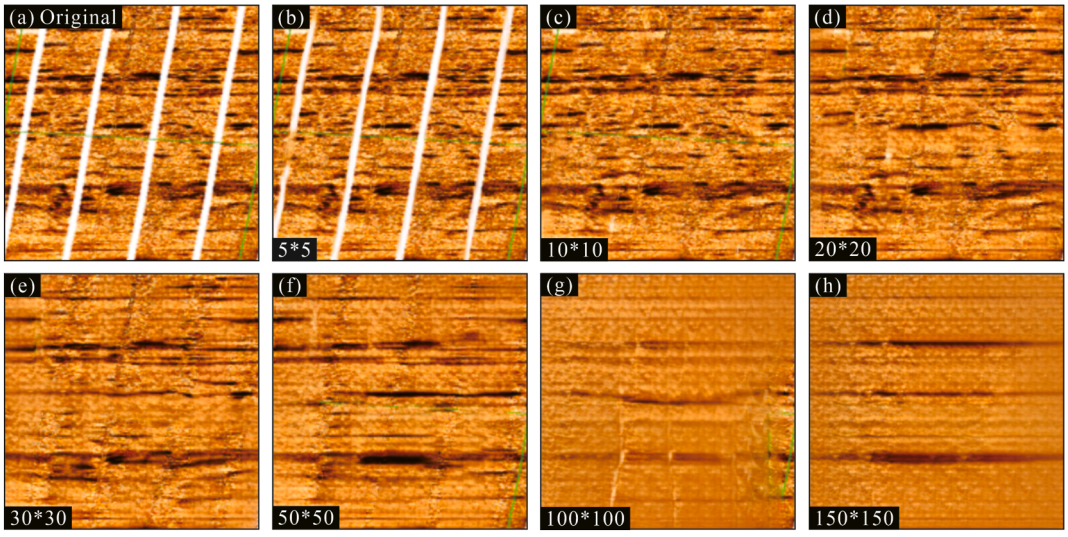

Fig. 10. DDPM performance with different window sizes: (a) Original; (b) 5×5 pixels; (c) 10×10 pixels; (d) 20×20 pixels; (e) 30×30 pixels; (f) 50×50 pixels; (g) 100×100 pixels; (h) 150×150 pixels.

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targets areas containing no more than one strip. This configuration ensures that the inpainting process addresses a single strip, thereby maintaining both local details and global uniformity, as shown in Fig. 10c–e. With increasing window sizes, the results tend to emphasize global uniform features (Fig. 10c–e). Ultimately, when the window size exceeds both the strip width and the inter-strip distance (Fig. 10f–h), the window encompasses at least two strips. The averaging effect over multiple strips can distort the results, retaining only the most prominent local high-frequency signals (Fig. 10f–h).

Window geometry represents another critical variable. Previous studies have established that square windows ranging from 20 to 30 pixels provide high-fidelity inpainting. To investigate the effect of geometric dimensions, we evaluated DDPM performance across different height-to-width (h/w) ratios, with the width fixed at 20 pixels (Fig. 11a–d). Visualization of HC-VR attention maps indicates that windows with larger h/w ratios cause the DDPM model to focus more on edges when the height exceeds the width (Fig. 11g and h). Conversely, square or near-square windows (Fig. 11e and f) may cause the model to overlook internal information, resulting in global averaging effects and reduced inpainting quality.

Lastly, the window movement strategy also affects the inpainting outcome. Inspired by convolutional operations in deep learning, the effects of stride variations were investigated. Previously, the stride was set to match the window size to avoid overlap. To examine the impact of overlap, results obtained with different strides were visualized (Fig. 12a–d), with the window size fixed at 20 pixels and strides set to 20, 10, 5, and 1. A stride of 20 indicates no overlap, whereas strides of 10, 5, and 1 represent half, quarter, and full overlap, respectively. As shown in Fig. 12a–d, increasing the degree of overlap results in greater local sharpness but a significant decrease in fidelity. This decline in image quality is attributed to the interaction of multiple information sources during the overlay process. Although multiple interactions can expand the receptive field of the HC-VR attention mechanism by capturing global sedimentary information, excessive overlap may cause the DDPM model to mistakenly treat unpainted image regions as original sedimentary information for new predictions, resulting in incorrect pattern stacking during image generation.

In summary, a square window is optimal for the DDPM inpainting process. The window size should exceed the width of the blank strips but remain less than or comparable to the

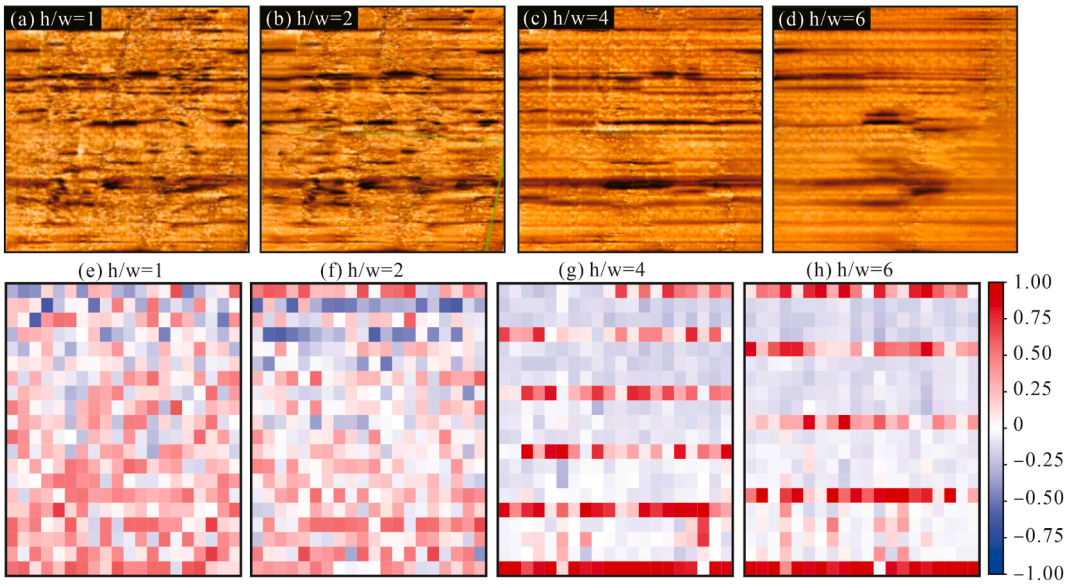

Fig. 11. DDPM performance and visualized HC-VR attention maps for different window height-to-width ratios. (a) Original; (b) Inpainted result, h/w = 2; (c) Inpainted result, h/w = 4; (d) Inpainted result, h/w = 6; (e) HC-VR attention map, h/w = 1; (f) HC-VR attention map, h/w = 2; (g) HC-VR attention map, h/w = 4; (h) HC-VR attention map, h/w = 6.

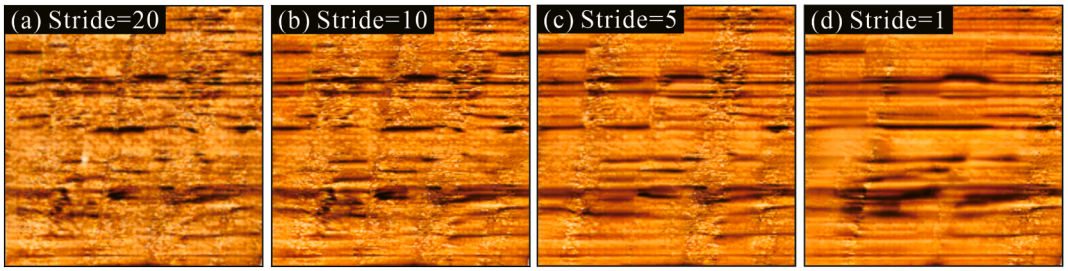

Fig. 12. DDPM performance under different stride settings: (a) Stride = 20 pixels; (b) Stride = 10 pixels; (c) Stride = 5 pixels; (d) Stride = 1 pixel.

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inter-strip distance. Regarding window movement, overlap should be avoided or kept to a minimum.

4.3. Geometric distortion effects of blank strips

The geometry of blank strips, particularly their width, is another critical factor influencing inpainting quality. As illustrated in Fig. 13, blank strips were generated as masks with widths ranging from 10 to 45 pixels on 250 × 250 pixel images. As the strip width increases, the inpainted FMI images exhibit greater detail loss in the intersection areas between the strips and image edges (dashed red frames, Fig. 13a2−h2). These regions present significant inpainting challenges because the DDPM model with HC-VR attention struggles to

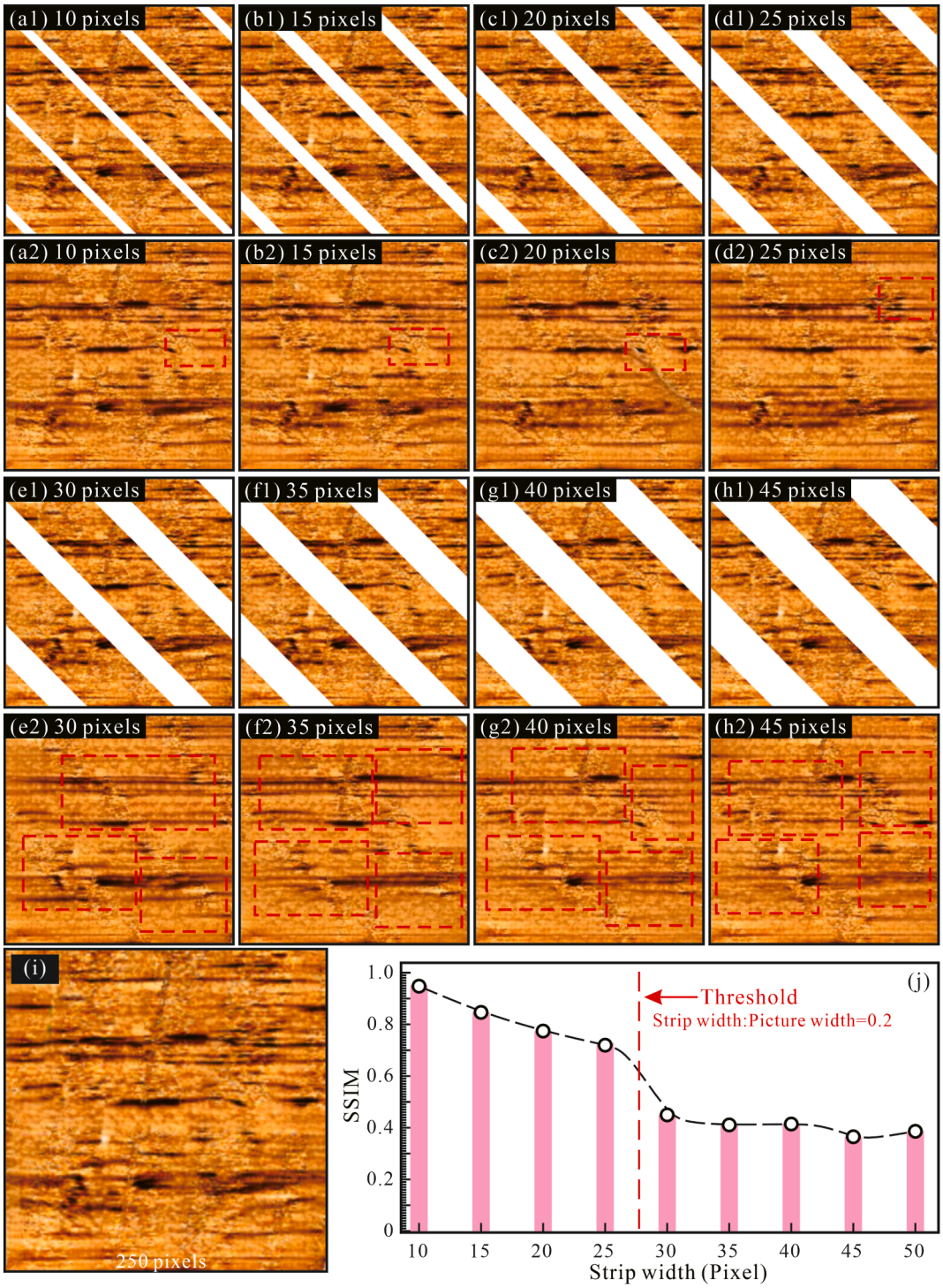

Fig. 13. Inpainted results with varying strip widths. (a) 10 pixels; (b) 15 pixels; (c) 20 pixels; (d) 25 pixels; (e) 30 pixels; (f) 35 pixels; (g) 40 pixels; (h) 45 pixels; (i) Original; (j) SSIM.

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capture adjacent depositional information required for repair. Specifically, increased strip width restricts the effective receptive field of the DDPM and signifies a greater loss of sedimentary information from the original image. Based on the ratio of strip width to image width, a threshold of 0.2 is established for inpainting FMI logs (Fig. 13j). In practical geophysical exploration, FMI logs with a blank strip-to-image width ratio below 0.2 are considered to have high repair potential. For the InpAS model, the lower limit for effective inpainting lies between 0.1 and 0.12 (Fig. 13j).

In actual FMI logging acquisitions, the width of blank strips is determined by the tool's pad configuration and borehole diameter rather than being arbitrary. Statistical analysis of the 184 m of FMI data used in this study revealed that blank strip widths range from 6 to 18 pixels, with a fixed image width of 250 pixels, and have a mean of 9.7 ± 2.3 pixels. The strip-width-to-image-width ratio predominantly ranges from 0.03 to 0.07, accounting for more than 80% of the sampled intervals. Extreme cases in which the ratio exceeds 0.2, as illustrated in Fig. 13j, are rare, accounting for less than 3% of the total footage, and typically correspond to severe borehole washouts in which the pad loses contact with the formation. Therefore, the proposed threshold of 0.2 effectively covers the majority of practical logging scenarios while defining the operational boundary for high-fidelity reconstruction.

Strip angles in FMI logs can vary considerably during hydrocarbon exploration. To simulate this variation, strip angles were rotated from 10° to 175°, representing a periodic symmetric rotation. Fig. 14 shows that as the strip angle approaches 90°, corresponding to a nearly horizontal orientation relative to the image edges, the repair effectiveness of the InpAS model decreases significantly, as indicated by the substantial reduction in SSIM (Fig. 14n). The inpainted FMI logs exhibit notable image distortion when the stripes are close to horizontal (Fig. 14f2−g2). When a blank strip is angled, a non-intersecting space always exists at the intersection of a square and a slanted rectangle. In the IoU calculation for object detection, such non-intersecting spatial information can maximize inpainting prompts. Consequently, the InpAS model has difficulty handling nearly horizontal bands in FMI logs, a challenge that generally complicates the repair of blank bands.

4.4. Analysis of artifacts

The inpainted results exhibit specific artifacts that reflect fundamental characteristics of the proposed methodology. As shown in Fig. 7, the slight blurring at object boundaries is characteristic of diffusion models, which inherently trade high-frequency sharpness for distribution diversity during iterative denoising. Unlike GANs, which may produce sharp but geologically inconsistent artifacts, the DDPM prioritizes statistical fidelity to the training distribution. This results in softer transitions that, while visually less crisp, maintain geological consistency with adjacent strata. In Fig. 10, the global averaging effects observed with large window sizes (50 × 50 and 100 × 100 pixels) result from excessive receptive fields encompassing multiple blank strips simultaneously. This violates the single-strip assumption underlying the local application of Walther's Law, causing the model to average features across disparate sedimentary intervals rather than reconstructing specific local geometries. As demonstrated, window sizes of 20−30 pixels optimally balance local detail preservation with global consistency. As shown in Fig. 14, the pronounced distortion when strip angles approach 90° (horizontal orientation) represents a fundamental theoretical limitation rather than an implementation failure. When strips are nearly horizontal, the vertical causal attention mechanism of HC-VR cannot access relevant lateral information from underlying strata. This violates the core assumption of Walther's Law that vertical superposition depends on horizontal adjacency, thereby confirming that the model correctly implements the geological constraint. Horizontal discontinuities cannot be predicted from underlying vertical sequences alone, highlighting the boundary conditions of the sedimentary knowledge-guided approach.

4.5. Analysis of influencing factors on the effectiveness of sedimentary knowledge

In addition to the geometric properties discussed above, the direction of sedimentary reference constitutes another critical factor influencing reconstruction quality. Although both bottom-up and top-down diffusion processes can mathematically reconstruct signal distributions from a purely statistical perspective, geological discontinuities, such as sedimentary hiatuses or erosional surfaces, disrupt signal correlation across boundaries. Fig. 15a shows the FMI log of a developed sedimentary discontinuity, where the dashed yellow line denotes the discontinuity. A sedimentary hiatus represents a gap in the geological record caused by non-deposition or erosion, which significantly impacts the interpretation of geological history and stratigraphic correlations. Geologically, sedimentary information near a discontinuity is primarily derived from the lower FMI image. Conversely, the sedimentary information above the discontinuity lacks geological correlation with the lower FMI image reconstruction and may even present contradictory sedimentary information. Fig. 15b shows a magnified view of the FMI log near the discontinuity. Fig. 15c and d illustrate the repainting results of the FMI log images containing sedimentary discontinuities, following bottom-up succession and top-down anti-succession, respectively. The repainting results following bottom-up succession exhibit clearer stratigraphic sedimentary structures (Fig. 15c). In contrast, the results following top-down succession display blurred and distorted sedimentary structures (Fig. 15d). From a signal processing perspective, DDPM modeling relies on the intrinsic relationships between input signals. On continuous sedimentary FMI logs, geological connections exist between different signal components, regardless of whether the processing follows succession or reverse succession. However, no such geological connection exists between signal segments separated by sedimentary discontinuities. Therefore, bottom-up reconstruction of FMI images, inspired by sedimentological knowledge, should serve as the fundamental approach for repainting blank strips.

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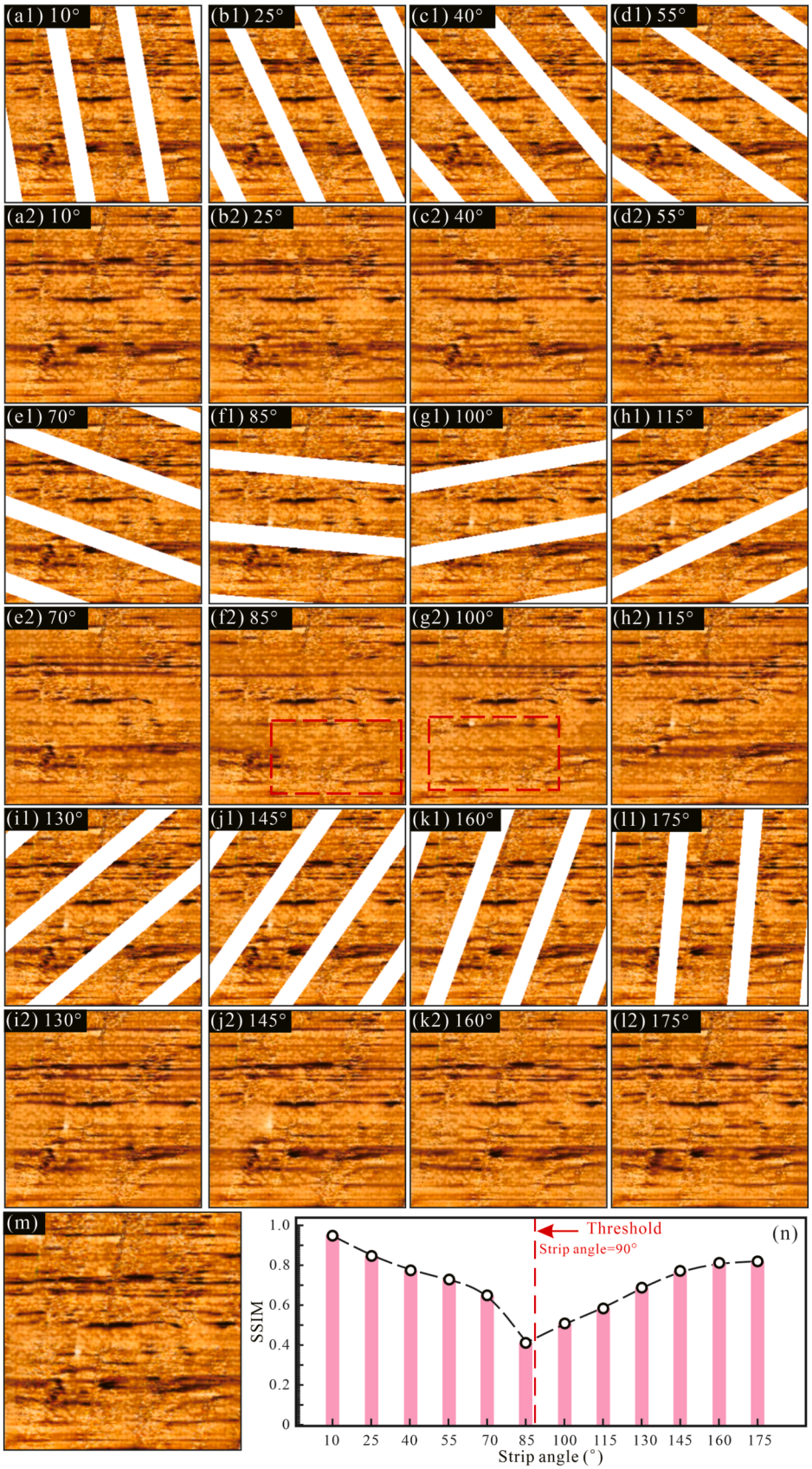

Fig. 14. Inpainted results obtained with varying strip angles. (a) 10°; (b) 25°; (c) 40°; (d) 55°; (e) 70°; (f) 85°; (g) 100°; (h) 115°; (i) 130°; (j) 145°; (k) 160°; (l) 175°; (m) Original image; (n) SSIM map.

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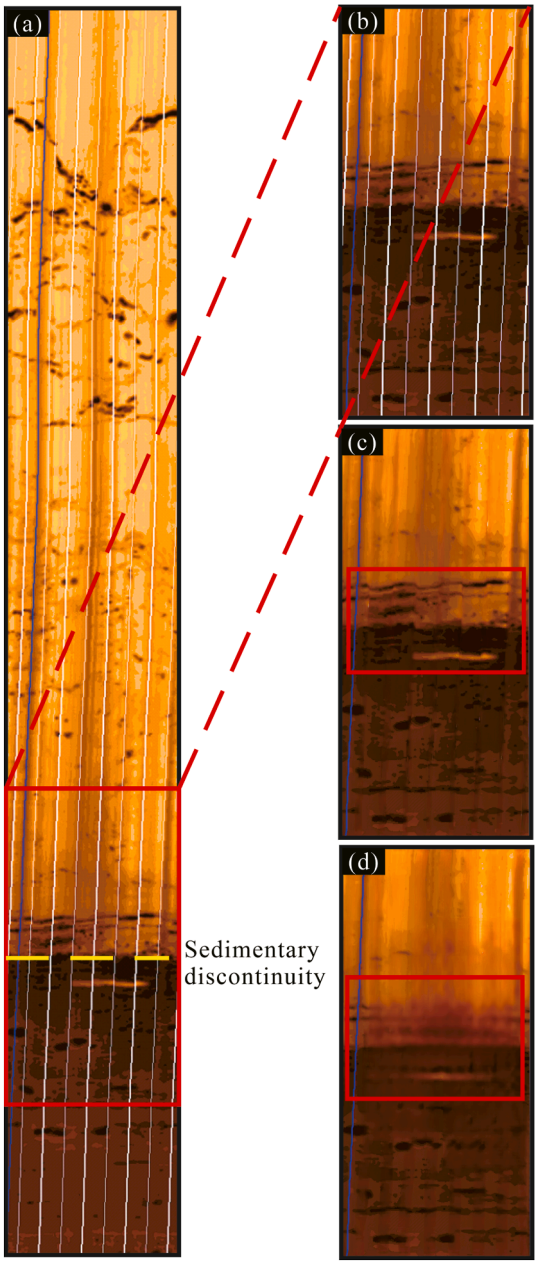

Fig. 15. Inpainted results for sedimentary discontinuities. (a) Original image; (b) Zoomed FMI log showing sedimentary discontinuity; (c) Inpainted result adhering to stratigraphic succession; (d) Inpainted result violating stratigraphic succession.

4.6. Limitations and future work

While the proposed approach demonstrates significant advancements, its limitations and directions for future research must be acknowledged. One limitation concerns the geological diversity of the validation dataset. The current experiments focus primarily on FMI images containing lamination and clot-like structures common in carbonate reservoirs, with limited representation of high-angle fractures, vuggy porosity, gravelly textures, and highly heterogeneous sedimentary architectures. Although the HC-VR DDPM was pretrained on large-scale diverse datasets (Place 2, Places 2, ImageNet, Open Images), suggesting potential for broader generalization, and the mathematical formulation based on Walther's Law theoretically extends to any vertically stratified system, explicit validation on these complex geological features remains a subject for future work. In addition, the approach relies heavily on the quality of input FMI data. Noisy or low-resolution images may reduce the detection accuracy of the F-SAM model, particularly in zones with poor borehole conditions where blank strip boundaries become indistinct. The HC-VR Attention mechanism, although effective for the tested succession, may not generalize to all geological formations without architectural adjustment, given the diverse and complex nature of sedimentary environments that may violate the stationarity assumptions underlying Walther's Law, such as mass transport deposits or chaotic facies patterns.

Although classical image processing techniques, such as Otsu thresholding and Canny edge detection, can efficiently and accurately identify high-contrast blank strips under ideal borehole conditions, a fine-tuned Segment Anything Model (F-SAM) was adopted to investigate a deep-learning-based approach that establishes an end-to-end framework from semantic-aware segmentation to geologically constrained image generation. However, F-SAM requires substantial computational resources because of its large number of parameters, and its supervised training paradigm differs fundamentally from the self-supervised mechanism used in the subsequent DDPM, preventing seamless joint training. Therefore, this approach is not intended to replace traditional methods. Future work will investigate lightweight adapter compression and the direct use of classical thresholding as a rapid preprocessing step, allowing the DDPM to focus on the core generation task. Regarding cross-geological applicability, the performance of InpAS was further evaluated across different lithologies, sedimentary structures, and drilling environments. Owing to the vertical causal attention mechanism constrained by Walther's Law, the model performs well in layered carbonate formations characterized by lamination and clot-like structures. However, in highly heterogeneous settings, such as fractured reservoirs, conglomerates, and breccias, the model has difficulty capturing high-frequency structural variations because representative depositional patterns are insufficiently represented in the training data. Specifically, when blank strips approach a horizontal orientation of approximately 90°, or occur in non-continuous depositional environments, such as sedimentary hiatuses and slump deposits, the rigid constraints imposed by the causal vertical mask may result in the loss of lateral information, substantially reducing restoration quality and introducing artifacts. In addition, reduced signal-to-noise ratios caused by borehole breakouts or mud invasion may impair the edge delineation accuracy of F-SAM, thereby affecting the conditional inputs for subsequent generation. Given these limitations, future work should establish a robustness testing framework based on different geological scenarios and develop restoration strategies tailored to specific lithologies and structural contexts. Integrating multimodal well-log data, such as conventional resistivity curves, could also provide supplementary

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geological constraints. Overall, InpAS currently provides an effective approach for FMI image restoration in stratified and continuously deposited environments. However, reliable performance in highly heterogeneous formations and under extreme drilling conditions will require further expansion of diverse training datasets and improvements in the adaptive capabilities of the algorithm.

Despite the promising results obtained for laminated and banded FMI textures, two primary limitations should be considered. First, the current dataset excludes complex lithologies, such as vuggy carbonates, fracture networks, conglomerates, and heterogeneous clastics. Therefore, the generalization of the model to these real-world geological settings remains unverified. This limitation reflects a deliberate "methodology-first" strategy aimed at validating the feasibility of incorporating Walther's Law into a diffusion attention mechanism under controlled stratified conditions. Future studies will expand the dataset, conduct systematic generalization assessments, and investigate domain adaptation and transfer learning. Second, drilling and acquisition factors, including mud type, borehole rugosity, tool eccentricity, and mud cake, have not been quantitatively assessed. Although the training data obtained from actual logs inherently include some degree of variability, the sensitivity of F-SAM detection and the robustness of HC-VR DDPM under degraded acquisition conditions remain unknown. Future studies will incorporate data augmentation to simulate variable acquisition conditions, develop dedicated preprocessing, and conduct benchmarking across diverse borehole environments to establish the operational limits of the proposed approach. The main contribution of this study is the development of a methodological framework demonstrating that sedimentological principles can be mathematically incorporated into a generative model for image restoration. Validation across a broader range of geological settings and adaptation to challenging acquisition conditions constitute the next major stages of research. Recognizing these limitations and defining a clear roadmap for future work provides a balanced assessment of the current capabilities and future potential of InpAS.

Future work should prioritize expanding the training and testing datasets to include fractured carbonates, conglomerates, and vuggy reservoirs, thereby comprehensively assessing generalization across diverse geological settings. Enhancing model robustness against varying data qualities through advanced denoising preprocessing or quality-aware adaptive mechanisms represents another critical direction. Furthermore, exploring adaptive window selection strategies that automatically adjust to local geological contexts, rather than relying on fixed-size windows, could improve reconstruction accuracy in heterogeneous formations. Integrating other geoscience principles and multimodal data, such as seismic profiles or conventional well-log data (e.g., gamma ray and resistivity curves), could provide additional stratigraphic constraints and further improve the geological consistency of the inpainting process. Another promising area involves developing unsupervised or weakly supervised learning techniques that require less annotated data, thereby reducing reliance on manual annotation for F-SAM training. Finally, investigating the transferability of the HC-VR Attention mechanism to other image restoration tasks in geosciences, such as seismic data interpolation or outcrop image repair, would demonstrate the broader applicability of sedimentary knowledge-guided deep learning architectures.

Despite promising performance on laminated carbonate reservoirs, applying InpAS to complex lithologies (e.g., fractured carbonates, conglomerates, breccias, and vuggy reservoirs) requires caution. Preliminary tests indicate that when sedimentary textures deviate significantly from the stratified continuity assumption (e.g., chaotic collapse breccias or high-angle fracture networks), the HC-VR attention mechanism, constrained by vertical causal masking, may fail to capture high-frequency spatial variations, leading to over-smoothing. For engineering deployment, users are recommended to first evaluate the textural heterogeneity of target formations. If the target interval exhibits pervasive fracturing or rubble textures, the restored images should be interpreted qualitatively rather than quantitatively.

Regarding computational overhead, the F-SAM module introduces significant inference latency due to its heavy ViT backbone (4.3 M learnable parameters). For scenarios with ideal borehole conditions where blank strips exhibit sharp, high-contrast edges (e.g., conductive mud systems with stable borehole diameters), a lightweight alternative is proposed: replacing F-SAM with classical Otsu adaptive thresholding [36] or Canny edge detection [37,38] coupled with morphological closing. A comparative benchmark is provided in the code repository. For poor-quality boreholes (rugose or washed-out sections), the full F-SAM remains recommended to ensure robust mask generation.

5. Conclusion

This study presents InpAS, a deep learning model guided by sedimentary knowledge for inpainting blank strips in FMI images. By explicitly incorporating Walther's Law into the HC-VR attention mechanism through causal vertical masking and horizontal facies correlation, the proposed approach addresses the challenge of reconstructing missing geological signals while adhering to stratigraphic principles.

The proposed method demonstrates that integrating domain knowledge (sedimentology) with generative models (diffusion) yields geologically consistent reconstructions, as supported by distribution-alignment metrics and PCA-based feature consistency (Fig. 9). In the 10-fold cross-validation, InpAS achieved superior performance among the evaluated statistical, sparse-reconstruction, CNN-, GAN-, score-, and diffusion-based baselines across FID, IS, SSIM, RP, PSNR, and MAE. These results indicate improved reconstruction of stratified FMI images, particularly in preserving lamination and clot-like textures. However, these conclusions are limited to the datasets, blank-strip configurations, and geological conditions tested in this study.

Nevertheless, the method has several limitations. Reconstruction quality decreases for near-horizontal blank strips,

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with angles approaching 90°, where the assumptions underlying Walther's Law become less applicable. In addition, the current validation focuses primarily on stratified carbonate reservoirs, with limited evaluation of fractured and vuggy formations. The diffusion-based trade-off between output diversity and edge sharpness also results in slightly blurred boundaries compared with GAN-based methods.

This study highlights the potential of integrating sedimentological principles into machine learning models for geological data analysis. InpAS bridges geophysics and deep learning by showing how geological principles, such as Walther's Law, can be mathematically integrated into neural network architectures to constrain generative processes. Future research will focus on extending the geological constraints to high-angle fractures and vuggy textures, developing adaptive window strategies, and incorporating multimodal well-log data to further improve geological consistency.

CRediT authorship statement

Xueni Xiang: Writing − review & editing, Writing − original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Shun Xia: Writing − review & editing, Supervision, Project administration. Jing Li: Designed the HC-VR Attention Mechanism. Keran Li: Designed the Combined Vertical and Horizontal Attention Mechanism. Wei Chen: Designed the Vertical and Horizontal Attention Mechanisms. Liang Zhao: Writing − review & editing, Supervision, Project administration. Jian Cao, Xun Yang, and Guobin Jiang: Contributed to coding. All authors reviewed the manuscript.

Data availability

Source code snippets are available at https://github.com/ai4geology/InpAS/tree/main. The data used in this research are available upon request by contacting the corresponding author at xias@petrochina.com.

Conflicts of interest

The authors declare the following financial interests and personal relations which may be considered as potential competing interests: Xueni Xiang, Jing Li, Liang Zhao, Guobin Jiang are currently employed by HSE and Technical Supervision Research Institute, PetroChina Southwest Oil & Gasfield Company. Shun Xia, Jian Cao, Xun Yang are currently employed by Northwest Sichuan Gas District, PetroChina Southwest Oil & Gasfield Company. Wei Chen is currently employed by China ZhenHua Oil Co., Ltd., Beijing 100037, China.

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